

Cell Datasheet - t7_r2 HEV cell

VBF-T7R2-DSH-001 | Cell Datasheet | t7_r2 HEV (R4_F1_Fine2_Tp05_Por32_H80)

Electrical:

1C discharge capacity	5.0438 Ah	(mass 42.4 g -> 119 Ah/kg)
Midpoint voltage	3.7919 V	
DC resistance	3.391 mOhm	
Power density	29066 W/kg	(calc-energy contract formula)

Energy:

Cell energy	17.98 Wh	
Gravimetric ED	423.62 Wh/kg	(contract: electrolyte excluded)
Volumetric ED	871.80 Wh/L	

Mechanical:

Stacking	pouch, area 0.1027 m ² , total thickness 200.8 um
Layer stack	pos 75.6 sep 12 neg 85.2 Al 16 Cu 12 (um)

Fast charge (protocol 4C_charge_45C = 1C discharge to 2.5 V then 4C CC to 4.2 V, 45 C):

anode interface potential min	+0.009007 V vs Li/Li+	(no lithium plating)
charge acceptance before 4.2 V cutoff	0.3844 Ah	(~69 s at 4C)
T_max during 4C	329.80 K	

Cycle life (aging_1C_100cyc_45C, isothermal SPMe + SEI ec-reaction-limited):

capacity cycle 1	0.8087 Ah	; cycle 100 0.7472 Ah (retention 92.4%)
SEI thickness end	463.14 nm	(criterion <= 550)

Abuse - nail penetration (10 W internal short-circuit, hA = 0.4248 W/K):

thermal runaway	triggered = False	; T_max 321.73 K ; dT/dt max 0.259 K/s
hot-soak probe (45 C start)	triggered = False	; T_max 321.73 K ; dT/dt 0.039 K/s

Basis: PyBaMM DFN/SPMe + bda calc-energy + bda run-tr; parameter set Chen2020

with deltas Chen2020 + {r_neg: 5.86->2.0 um; t+ 0.2594->0.5; negative porosity 0.25->0.32; h 10->80 W/m²/K}.